

ERIC HOLT

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PROFESSIONAL SUMMARY

Full-stack AI engineer who builds production-grade systems including agentic pipelines, LLM evaluation platforms, automated delivery services, RAG pipelines, multi-agent orchestration, and MCP-integrated tools, deployed to real infrastructure with observability, testing, and CI/CD. Currently leading a full-stack migration of a production AI assistant from Azure AI Foundry to a LangGraph/Claude architecture at MarineMax, including a claude.ai-parity feature build and a self-directed security hardening pass.

TECHNICAL SKILLS

Programming: Python (Certified), JavaScript, TypeScript, HTML/CSS, C#

AI/ML Frameworks: LangGraph, PyTorch, TensorFlow, Transformers, Scikit-Learn

Agent & LLM: Groq inference, RAG pipelines, vector search, pgvector, LLM fine-tuning (LoRA), agentic orchestration, LLM-as-judge evaluation, MCP (Model Context Protocol), FastMCP, reinforcement learning, Anthropic API, Azure AI Foundry, event-driven webhook ingestion, HMAC-SHA256 webhook verification

Web Development: FastAPI, Node.js, Express, React, Vite, Tailwind CSS, Flask, REST APIs, WebSocket, SSE, GraphQL

Data & Analytics: PostgreSQL, NumPy, Pandas, Recharts, SQL analytics, asyncio, Snowflake/Cortex

Tools: Git, Docker, Docker Compose, AWS (EC2, RDS, CloudWatch), Supabase, Neon, pgvector, fastembed, Stripe, Gmail OAuth2, Resend, GitHub Actions, Azure, Entra ID

TECHNICAL PROJECTS

TraceAgent — Observable Agentic Research Platform | Python, FastAPI, Groq, Tavily, React, PostgreSQL, Docker, AWS

- Built a custom agentic pipeline with four sequential stages — planner, search, reflection, and synthesis — where the reflection stage evaluates result sufficiency and conditionally loops back with refined queries; every internal decision is persisted to PostgreSQL and streamed to the browser in real time via WebSocket, with run forking with lineage tracking and side-by-side run comparison.
- Deployed to AWS (EC2 t3.micro + RDS PostgreSQL on private VPC subnet) and Hugging Face Spaces; instrumented with CloudWatch Logs via watchtower and 6 custom CloudWatch metrics per run feeding a live pipeline dashboard; pytest integration test suite covers route lifecycle, pipeline step verification, and fork lineage; GitHub Actions CD pipeline auto-deploys to both Hugging Face Spaces and AWS EC2 on every push.

DailySignal — Automated AI News Briefing Service | Python, GitHub Actions, Groq, PostgreSQL, Neon, Resend, React, Recharts, Docker

- Built a fully automated twice-daily AI news briefing service — zero manual steps from fetch to inbox; GitHub Actions cron pulls from HackerNews, Product Hunt GraphQL, and five RSS feeds, deduplicates articles via SHA-256 content hashing, synthesizes a structured briefing via Groq, and delivers to subscribers via Resend with per-subscriber unsubscribe tokens and automatic deactivation after 3 failed deliveries.
- GitHub Actions as serverless infrastructure — no always-on server required; PostgreSQL audit log per subscriber per run; 55 unit tests at 98% coverage; deployed to Hugging Face Spaces with Neon serverless PostgreSQL; FastMCP server mounted at /mcp exposes briefings, run history, and subscriber counts as tools for AI agents.

EvalBoard — Systematic LLM Evaluation Platform | Python, FastAPI, Groq, PostgreSQL, React, Docker, GitHub Actions

- Built a platform for measuring LLM output quality in a repeatable, structured way — define test cases with expected outputs and scoring criteria, run them against one or two models simultaneously, and watch scores stream to the browser live via SSE as each case completes; an LLM-as-judge second Groq call returns structured JSON per case: score (1-10), pass/fail, strengths, weaknesses, and reasoning.
- Fan-out SSE broadcaster supports multiple simultaneous clients with missed-event replay on reconnect; prompt variant versioning enables A/B testing system prompts against the same suite; dashboard tracks score trends, model leaderboard, and per-suite pass rates with side-by-side run comparison; async SQLAlchemy with Alembic migrations; deployed to Hugging Face Spaces via Docker with GitHub Actions CI/CD.

SupportDesk — AI-Powered Customer Support Automation | Python, FastAPI, Groq, pgvector, React, PostgreSQL, Docker, GitHub Actions

- Built a webhook-driven support automation pipeline — HMAC-SHA256 verified ticket ingestion, Groq LLM classification (type, urgency, confidence), pgvector RAG retrieval against a knowledge base, and SSE-streamed reply drafts delivered to agents before a ticket is opened; agents review cited sources inline, approve, and Resend delivers the reply; every pipeline stage broadcast to a live WebSocket event dashboard.
- HMAC-SHA256 webhook verification with timestamp replay protection; fastembed + pgvector cosine similarity RAG retrieval; human-in-the-loop approval gate before any reply is sent; 71 pytest tests; deployed to Hugging Face Spaces via Docker with GitHub Actions CI/CD.

TriageAgent — AI-Powered Email Triage & Draft System | Python, FastAPI, Groq, Gmail OAuth2, React, Docker, GitHub Actions

- Built a three-stage agentic pipeline — classify, extract, draft — that processes pasted emails end-to-end: classifying type and urgency with a confidence score, extracting sender intent and action items, then generating a professional reply streamed to the browser in real time via SSE; results below 0.75 confidence are flagged for human review before sending.

- Gmail OAuth2 integration sends the approved draft directly from the UI; confidence threshold safety gate enforces human-in-the-loop approval before any action; deployed as a Dockerized service to Hugging Face Spaces with automated CI/CD via GitHub Actions; pytest test suite with separate unit (fully mocked) and integration layers covering all three pipeline stages.

RepoSignal — GitHub Issues Analytics Dashboard | Python, FastAPI, Groq, GitHub API, PostgreSQL, React, Recharts, Docker

- Built a full-stack analytics platform that fetches real issues from any public GitHub repo via REST API, classifies each by type, priority, and sentiment using a parallelized LLM pipeline (asyncio semaphore, 15 concurrent calls), and persists results to PostgreSQL across three normalized tables; SQL-powered dashboard uses GROUP BY, JOIN, DATE_TRUNC, and window functions to drive five live charts including a weekly time-series view.
- Filterable issue table with type/priority/status controls; SSE-streamed AI triage draft for any open issue with one-click clipboard copy; repo history with one-click reload from DB; 9 API routes with full pytest unit (fully mocked) and integration test coverage; multi-stage Docker build deployed to Hugging Face Spaces with Neon serverless PostgreSQL; FastMCP server mounted at /mcp exposes 3 read-only SSE tools (list runs, get dashboard, get issues) queryable by any MCP-compatible AI client.

AgentMesh — Multi-Agent Code Review & Repair System | Python, LangGraph, FastAPI, Groq, React, Tailwind CSS, Docker

- Built three specialized agents collaborating via a LangGraph StateGraph: Reviewer identifies issues, Fixer produces corrected code with a changelog, and Evaluator scores the fix across correctness, completeness, and code quality — rejecting and routing back to Fixer with specific feedback for a second pass when the score falls below threshold.
- Every agent decision and handoff streams to the browser in real time via SSE; structured Pydantic schemas enforce all inter-agent state, with language auto-detection, multi-stage Docker deployment to Hugging Face Spaces, and a pytest test suite covering agent routing and evaluator scoring logic.

EXPERIENCE

AI Engineer — MarineMax May 2026–Present Full-Time / Remote

- Leading a from-scratch migration of PropellerAi, MarineMax's production multi-agent sales assistant, off Azure AI Foundry orchestration onto a LangGraph pipeline calling Claude directly via the Anthropic API — the largest engineering effort in the codebase's history (203 files changed, +42,570/-1,952 lines) — including cost-aware model routing (Haiku 4.5 for routing vs. Sonnet 5 for specialists), Azure Key Vault provisioning, and auth redesign for tool calls.
- Built full claude.ai feature parity into the migrated assistant, including Projects with cross-specialist file attachments, an Artifacts gallery with in-place PDF/DOCX preview, a from-scratch document-editing pipeline, voice dictation, RAW image handling, chat export, and Anthropic prompt caching.
- Ran a self-directed hardening pass that found and fixed 8 confirmed concurrency, correctness, and security bugs plus one bonus fix, and closed 7 test-coverage gaps before the migration reached customers.
- Built propeller_tools, a FastAPI microservice integrating Snowflake/Cortex, Bing web search, and a third-party marine valuation API behind an Azure AI Foundry tool contract, powering J.D. Power book-value and fair-market-value estimates for the sales assistant; owns its CI pipeline and 46-test suite.
- Top committer on PropellerAi (170 commits / 80 merged PRs) and sole contributor on BoatValueIntelligence's valuation microservice (84 commits / 34 merged PRs); triaged and remediated MarineMax's external CYPFER penetration test findings, including a critical LFI/path-traversal vulnerability and an unmasked-credentials leak via a health-check endpoint, both fixed in production the same week.

Software Engineer / Full-Stack Developer — Get Set Wedding Dec 2025–Feb 2026 Startup / Contract

- Shipped production features including auth/onboarding flows, Stripe billing enforcement with webhook-driven server-side plan gating, PostgreSQL RLS architecture, RBAC admin routing, and refresh-safe session management.
- Built a context-aware AI assistant with live access to user planning data including budget, guest list, and vendors for accurate personalized responses, alongside admin tooling and subscription-ready infrastructure.

Technical Diagnostics & Systems Specialist — Emergency Vehicle Platforms 2017–2024 City of Coppell • City of Grapevine • Siddons-Martin Emergency Group

EDUCATION

DeVry University — B.S. in Software Development Expected 2028 A.S. in progress, expected 2026 • Concentration: AI and Data Analytics

CERTIFICATIONS & ACCOMPLISHMENTS

PCEP — Certified Python Programmer (2025) • Undergraduate Certificate in Programming Essentials, With Honors — DeVry University (2026) • Introduction to Model Context Protocol — Anthropic (2026) • Model Context Protocol: Advanced Topics — Anthropic (2026) • Claude with the Anthropic API — Anthropic (2026) • Dean's List, DeVry University (2026) • National Society of Leadership and Success — Inducted Member